Game of Hex and Multi-Armed Bandits

Bachelor's in Computer Science — Software Design 2

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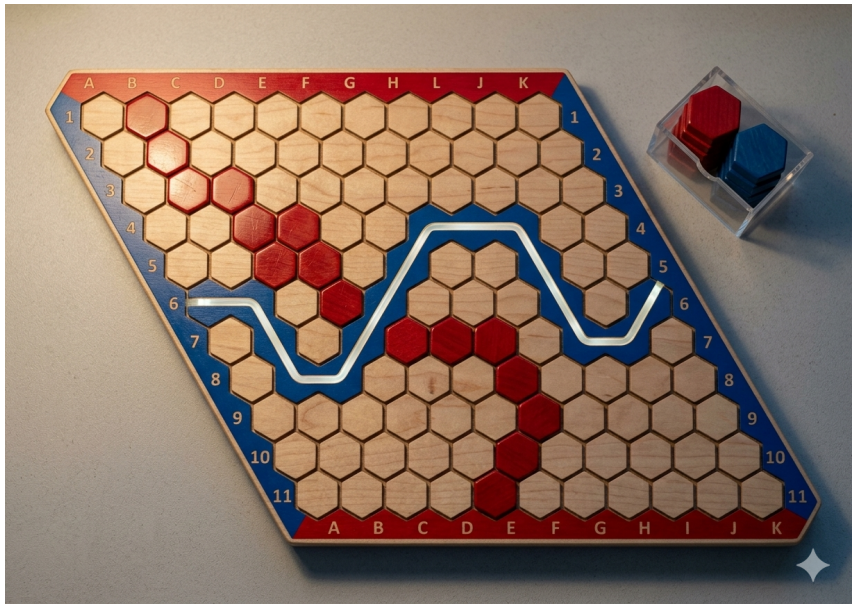
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Presentation Outline

1. Introduction
2. Objectives
3. Architecture
4. Implementation
5. Reverse Hex
6. Experiments
7. Conclusion

Introduction : The Game of Hex

- **Objective :** Connect the two sides of your color with an unbroken chain.
- **Property :** A perfect information game where a draw is impossible.



Problem Statement

How does the **budget ratio** between the two players influence their win probability relative to the board size?

- Is there a **budget threshold** beyond which a player becomes significantly advantaged?
- What is the impact of the first-player advantage on the win rate?

Project Objectives

- Implement the Game of Hex with robotic players:
  - Represent the board as a 2D grid
  - Win condition algorithm (**DFS**)
  - Random-move robot player
- Implementation of the **MCTS** algorithm
- Implementation of the **RAVE** algorithm
- Implement the **Reverse Hex** version
- Performance analysis through a series of experiments varying game configurations

Architecture — Package Organization

Package Game

- Board.java
- Player.java
- Robot.java
- RobotMCTS.java
- RobotRave.java
- DFS.java
- Position.java
- Game.java

Package mcts

- MCTS.java
- NodeMcts.java
- State.java
- CacheMap.java

Package Rave

- RAVE.java
- NodeRave.java
- Simulation Result.java

Reverse Hex

Same packages — only `isAWin()` is inverted in `Board.java`

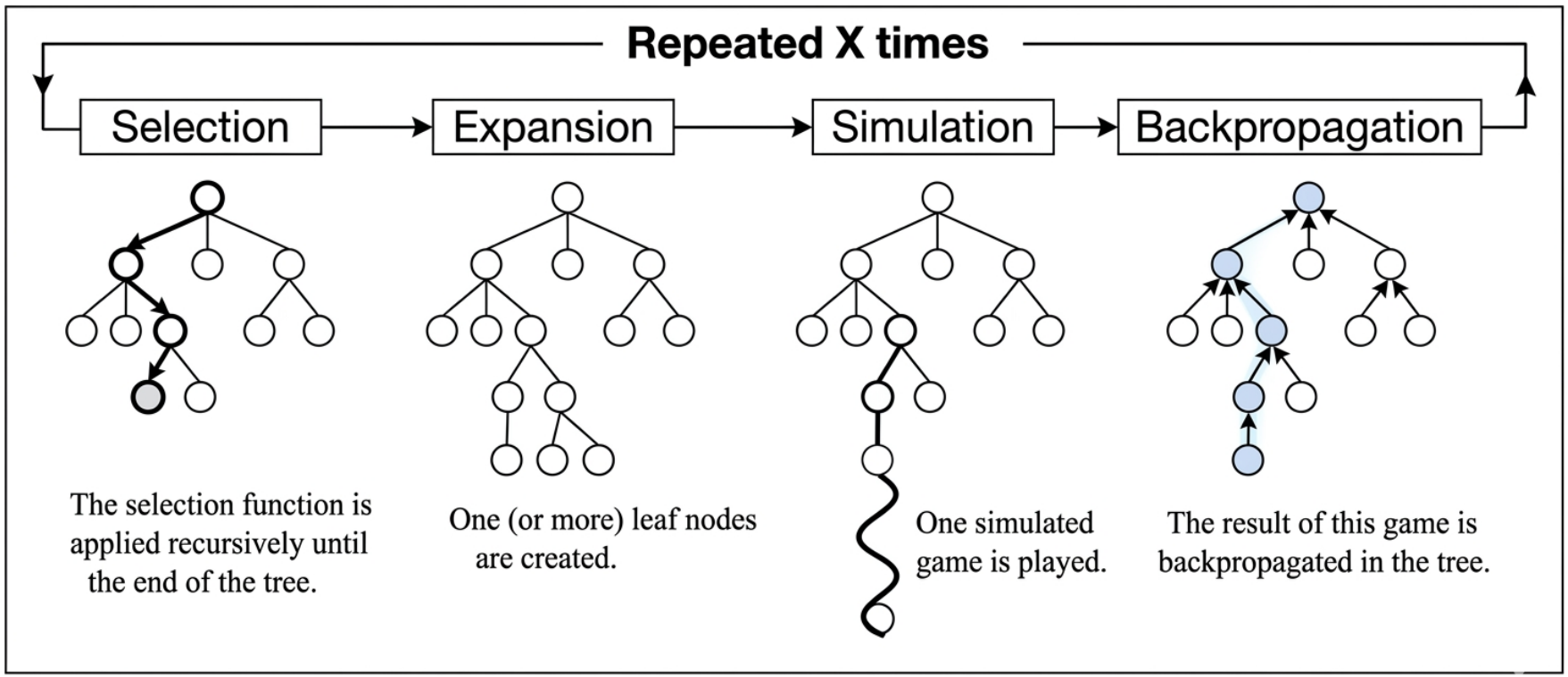
MCTS — Concept

What is MCTS?

A search algorithm guided by **random simulations**, without prior knowledge of the game.

- Unlike **Minimax** — no exhaustive exploration
- Focuses efforts on the **most promising moves**
- Each node in the tree represents a **game state**
- Each node stores: wins w_i and visits n_i

MCTS — The 4 Phases



Upper Confidence Bound for Trees (UCT)

Formula

$$\text{UCT}(i) = \frac{w_i}{n_i} + c \cdot \sqrt{\frac{\ln(n_p)}{n_i}}$$

- w_i: number of winning simulations for the current node
- n_i: total number of simulations for the current node
- n_p: total number of simulations for the parent node
- c: exploration parameter (typically $c = \sqrt{2}$)

Project Progress

- Phase 1: Setting up game basics, board, and DFS algorithm
- Phase 2: Implementing MCTS with all 4 phases
- Phase 3: Implementing RAVE and integrating the Reverse Hex variant
- Phase 4: Experiments with simulation configurations, result analysis, and final report

Why RAVE?

- RAVE is an optimization of the MCTS algorithm.
- The principle of RAVE is based on the AMAF (All Moves As First) heuristic.
- AMAF assumes that every move played in a simulation has a constant value, regardless of when it is played in the game.
- This allows for gathering statistics on all moves played in each simulation, which accelerates algorithm convergence.

RAVE Selection Formula

- $Q_{\text{RAVE}} = (1 - \beta) \cdot \frac{w_i}{n_i} + \beta \cdot \frac{\text{raveVictories}}{\text{raveVisits}} + C \cdot \sqrt{\frac{\ln(n_p)}{n_i}}$
- Where $\beta = \sqrt{\frac{k}{3 \cdot n_i + k}}$ and $C = \sqrt{2}$
- β decreases as n_i increases: a progressive transition from AMAF to MCTS.

Tree Update (MCTS and RAVE)

After each simulation, two updates are performed during **backpropagation**:

- **Classic MCTS Update**:
  - `victory()` and `visited()` for each ancestor node.
- **RAVE Update**:
  - `updateRaveStats()` to propagate AMAF results to ancestor nodes.
  - `SimulationResult` contains the result and the list of moves played, used for the AMAF update.

Concrete Result: With $k = 300$, **AMAF dominates** up to approximately 100 visits per node, then **MCTS** progressively takes over.

Reverse Hex

- Inverted Rule: Lose by connecting your own sides
- Defensive Strategy: Force the opponent to connect
- Minimal Modifications: inverted `isAWin()`, adapted simulation

Experimental Protocol

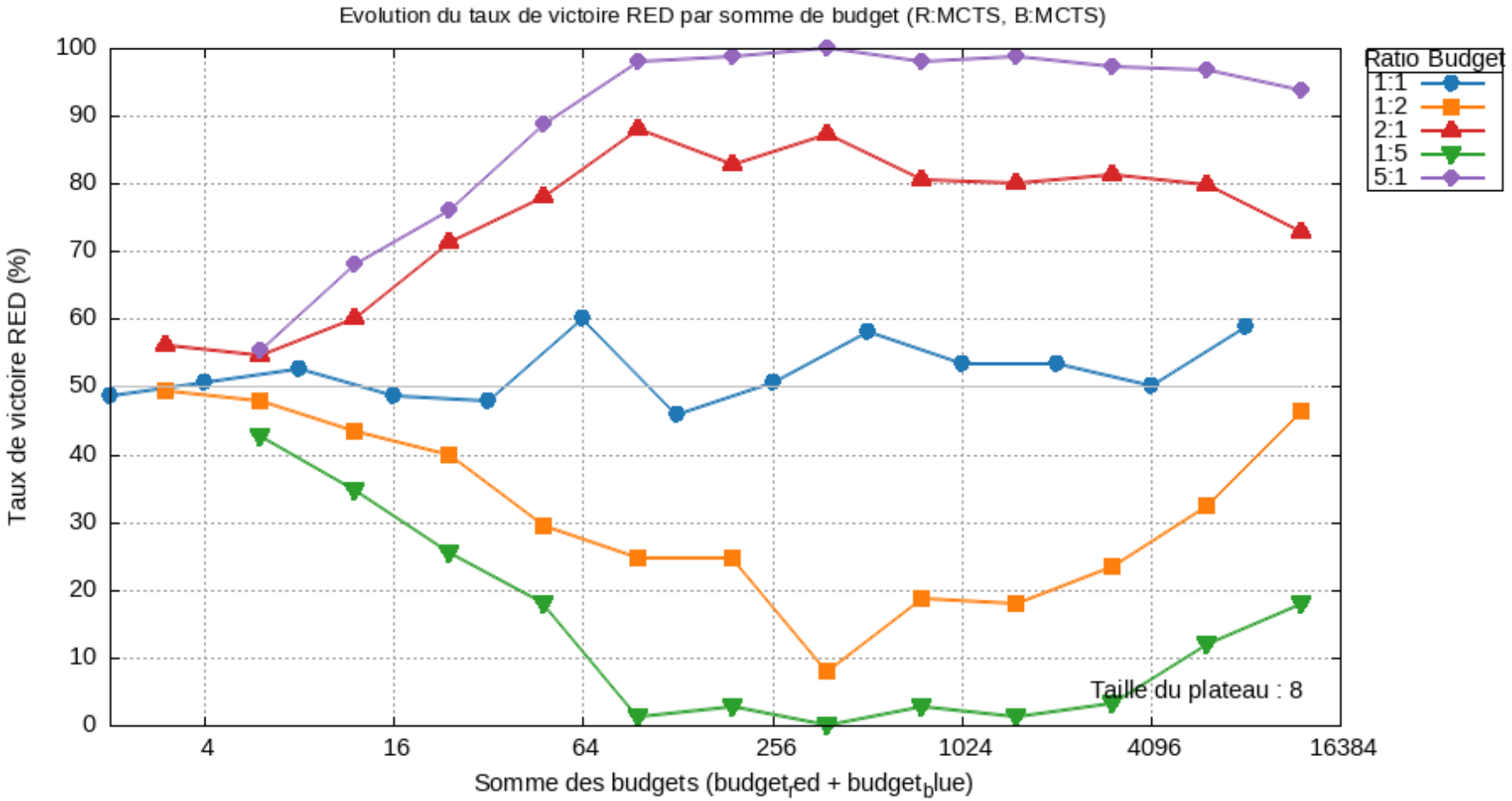
Defining the Budget

Budget = number of simulations **per move** (`getBestMove`). X-axis = sum of budgetRED + budgetBLUE per move.

Parameters

Algorithms	MCTS AND RAVE
Sizes	$4\times4 \cdot 8\times8 \cdot 10\times10$
Ratios	1:1 · 1:2 · 2:1 · 1:5 · 5:1
Games/Point	150 games
First Player	Always RED
RAVE Parameter	$k = 300$ (threshold $\approx$ 100 visits/node)

MCTS vs MCTS — 8×8 Board



MCTS vs MCTS — Analysis (8×8 Board)

- **Ratio is the dominant factor** — the hierarchy of curves is entirely dictated by the budget ratio between the two players
- **The first-player advantage, visible on size 4, becomes non-significant** — board complexity dilutes it
- **A critical threshold exists** — between ratios 1:2 and 1:5, the behavior changes in nature
- **At very high budgets, even a player with an unfavorable ratio accumulates enough simulations to converge toward optimal play**

MCTS vs MCTS — Generalization to other sizes

Distinct regimes based on board size.

4×4 Size

Ratio effects diminish at high budgets — both players eventually converge toward optimal play.

$\Rightarrow$

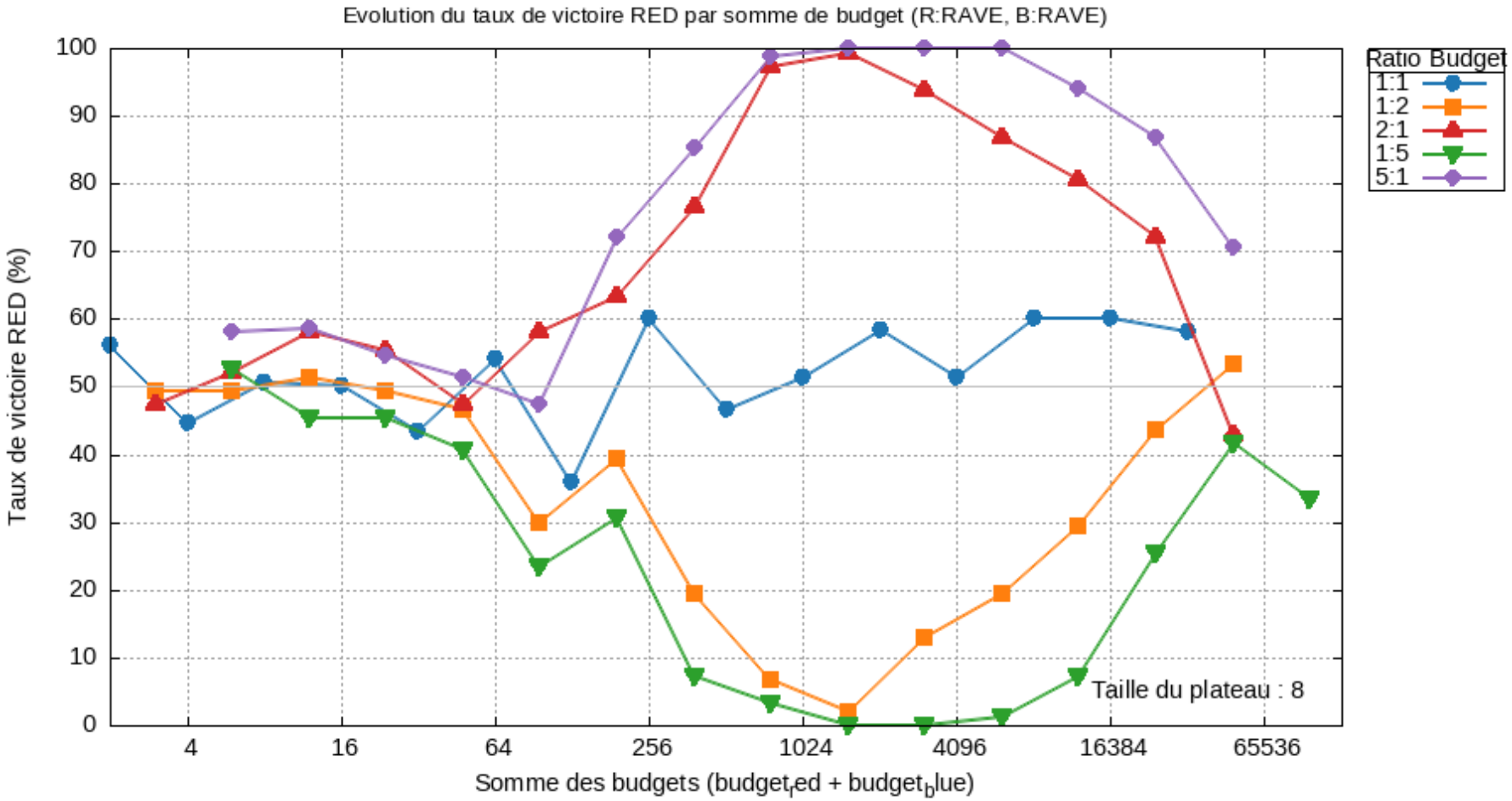
10×10 Size

Everything triggers earlier. The first-player advantage is completely absorbed by complexity.

Conclusion

The larger the board, the harder a budget disadvantage is to overcome — but at very high budgets, players converge toward optimal play.

RAVE vs RAVE — 8×8 Board



RAVE vs RAVE — Analysis (8×8 Board)

- **At low budgets, RAVE behaves like MCTS** — AMAF is not yet reliable
- **At intermediate budgets, RAVE outperforms MCTS** — it achieves the same levels with fewer simulations
- **At very high budgets, both converge** — $\beta \to 0$, MCTS takes back control
- **RAVE strengthens the first-player advantage** thanks to the AMAF bias on opening moves

RAVE vs RAVE — Generalization to other sizes

Same structure as with MCTS, but RAVE reaches the same performance levels with fewer simulations thanks to AMAF.

4×4 Size

AMAF statistics too sparse — unstable behavior. RAVE amplifies the first-player advantage via AMAF bias.

$\Rightarrow$

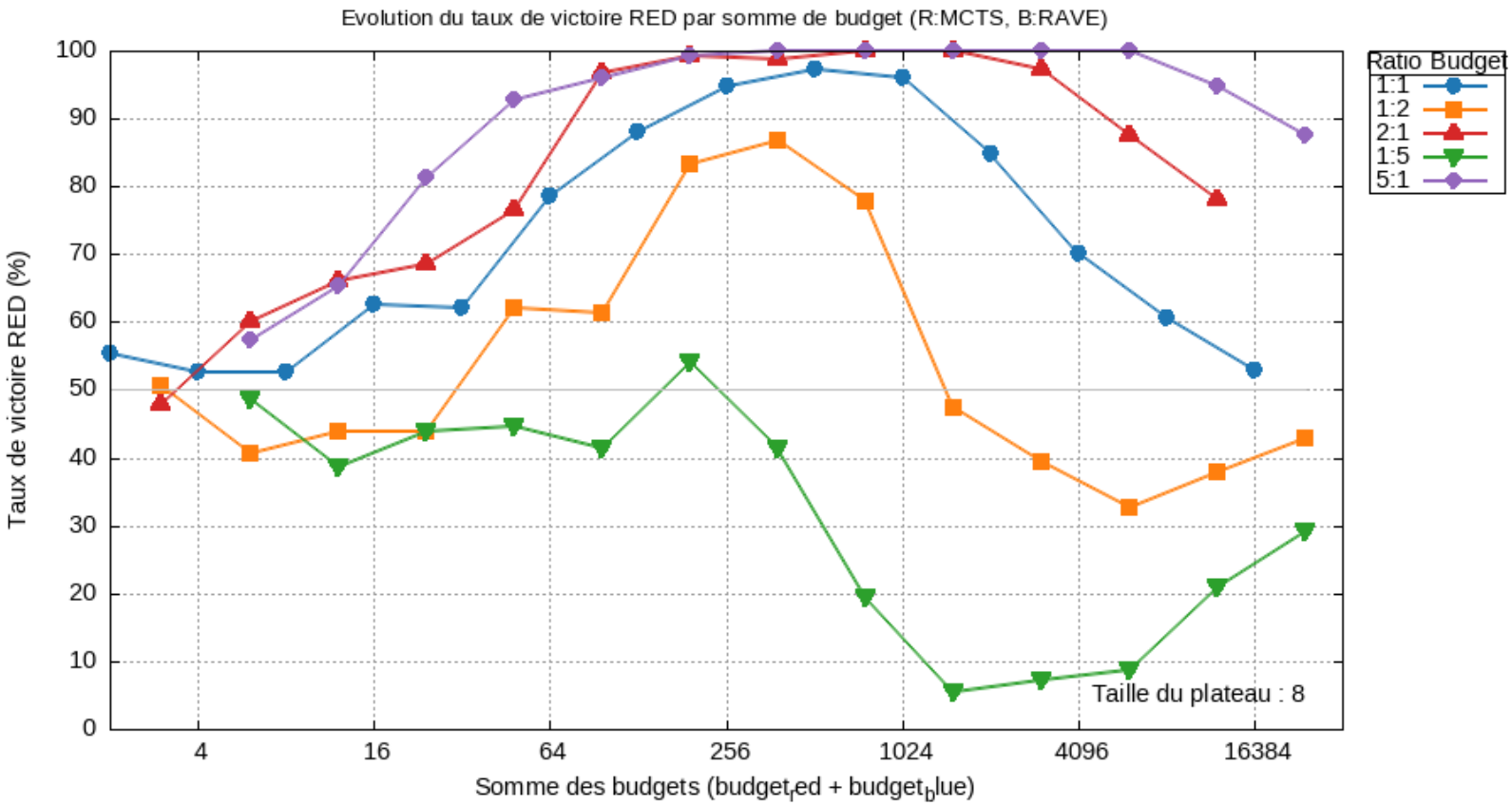
10×10 Size

Long simulations — AMAF operational very early. Most intense effects of the entire experiment.

Conclusion

The larger the board, the earlier RAVE becomes operational and the more intense its effects.

MCTS vs RAVE — 8×8 Board



MCTS vs RAVE — Performance Analysis (8x8)

- **RAVE as a blocking opponent** — In the second-player position, RAVE uses its global intuition to identify strategic cells and slow down the first player. It acts as a wall but fails to erase the lead advantage.
- **Correlation between win rate and budget** — Positive quasi-linear correlation. Budget allows MCTS to bypass RAVE's defense. Every doubling of the budget reduces uncertainty across the 64 cells.
- **Resilience against adverse ratios** — Even at a 1:5 ratio, the curve rises at high budgets. The first-move advantage eventually compensates for the computational deficit once MCTS reaches sufficient precision.

MCTS vs RAVE — Generalization to other sizes

Analysis of asymmetry based on board dimension.

4×4 Size

Low complexity allows MCTS to quickly find bypass paths. Both players quickly converge to optimal play.

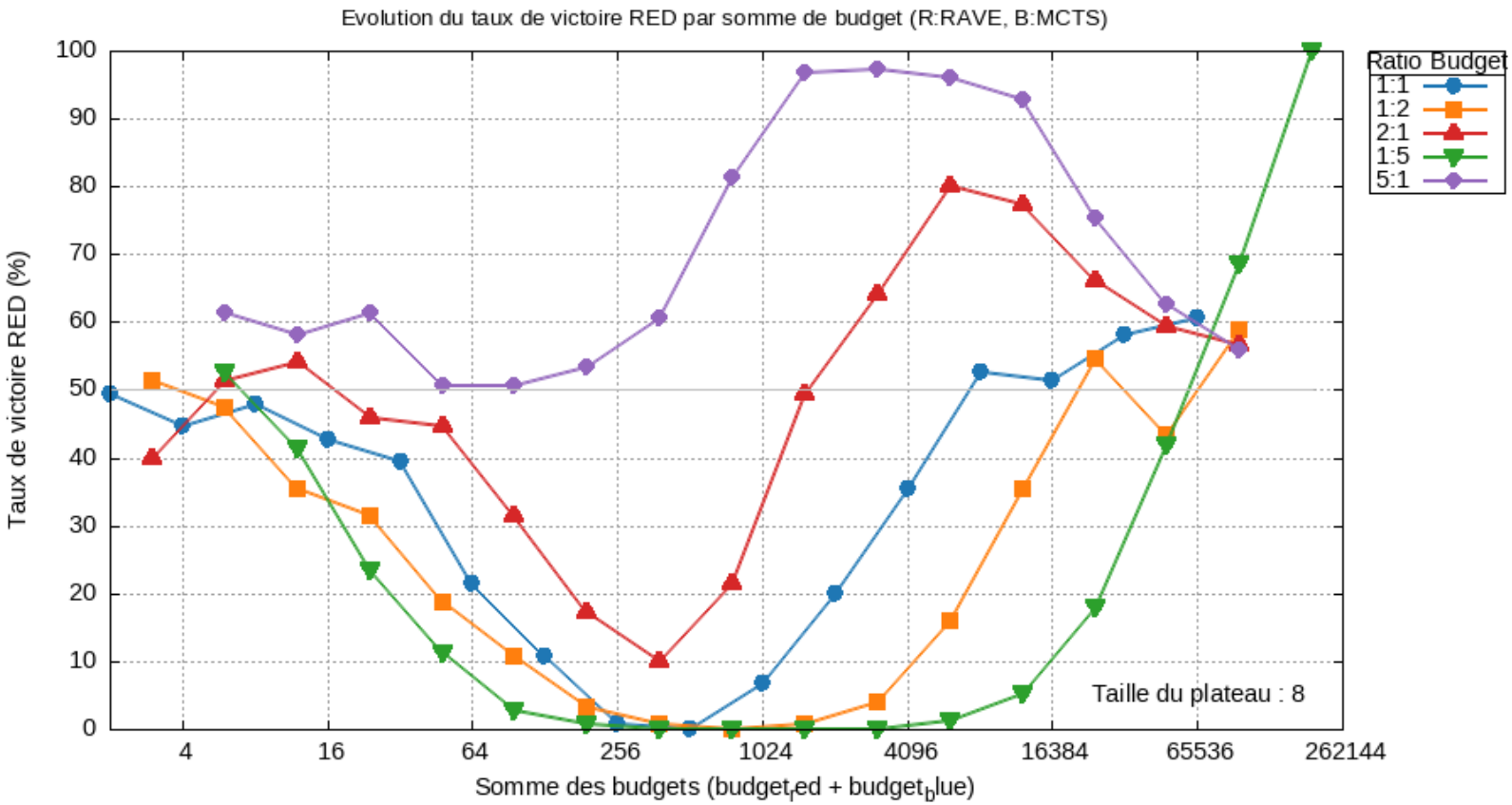
10×10 Size

The role of RAVE as a "blocker" is amplified. The first-player advantage is harder to achieve because complexity absorbs attack attempts.

Conclusion

The larger the board, the more effective the opponent's defensive role, requiring a massive budget to restore the lead advantage.

RAVE vs MCTS — 8×8 Board



RAVE vs MCTS — Performance Analysis (8x8)

- **RAVE as first move: The reckless attacker** — RAVE's aggressiveness pushes it to take huge risks. It "spreads out" too quickly on the board, leaving tactical gaps that MCTS (Blue) exploits to cut its thin lines.

- **Correlation between win rate and budget** — The first-player advantage is erased here at medium budgets by RAVE's mistakes. Budget efficiency only becomes profitable after 1024 simulations, where AMAF finally stabilizes the strategy.

- **Resilience against adverse ratios** — Despite a 1:5 disadvantage, RAVE eventually recovers. Its generalization capacity eventually corrects its initial aggressiveness to transform the first move into a systematic victory.

RAVE vs MCTS — Generalization to other sizes

Impact of heuristic aggressiveness according to board size.

4×4 Size

Restricted space limits RAVE's tactical errors. The AMAF bias immediately reinforces the first-player advantage.

10×10 Size

AMAF becomes operational very early. The "reckless attacker" effect is magnified by the length of simulations on large boards.

Conclusion

The larger the board, the more intense and early RAVE's effects (both positive and negative) become.

Synthesis

Research Question

What influence does the budget ratio have on the win probability depending on the grid size?

- **Budget ratio** $\rightarrow$ dominant factor
- **Large board** $\rightarrow$ amplified effects
- **RAVE** $\rightarrow$ faster convergence
- **First-player advantage** $\rightarrow$ secondary

Conclusion

Merci Pour Votre Attention !